Chain-of-Cooking: Cooking Process Visualization via Bidirectional Chain-of-Thought Guidance

Taro Yamada

Abstract

Cooking process visualization aims to generate a sequence of images depicting the intermediate states of a recipe, step by step. Unlike single-dish food image generation, this task demands tracking evolving ingredient appearances across steps while preserving procedural coherence. Existing approaches either focus on final-dish synthesis or treat each step independently, resulting in semantic drift and visually inconsistent sequences. We propose **Chain-of-Cooking (CoCook)**, a diffusion-based framework that addresses these challenges through three complementary modules: (1) a Dynamic Patch Selection module that retrieves text-matched visual patches from historical frames and injects them via a learnable visual attention path, (2) a Semantic Evolution module that initializes latent prompts from the previous step and cross-attends with the current text to propagate step-aware cues, and (3) a Bidirectional Chain-of-Thought Guidance mechanism that enforces semantic consistency across both forward and backward diffusion passes. We further introduce **CookViz**, a publicly available benchmark comprising 40,362 high-resolution, step-aligned image–text pairs for cooking process evaluation. Extensive experiments on CookViz and RecipeQA demonstrate that CoCook achieves state-of-the-art performance on all automatic metrics and human evaluations, establishing a new standard for coherent cooking process visualization.

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1 Introduction

The visual documentation of cooking processes plays an increasingly important role in culinary education, recipe sharing platforms, and food-related content creation. While a single photograph of a finished dish conveys the end result, it fails to capture the rich procedural transformations that ingredients undergo during preparation—the browning of onions, the thickening of a sauce, or the layering of a pastry. Generating a faithful visual sequence of these intermediate states from textual recipe instructions is the goal of *cooking process visualization*, a task that is fundamentally more challenging than conventional food image generation [15, 30, 37].

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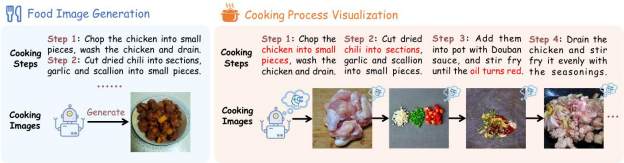

Figure 1: Comparison of traditional food image generation (left), which produces a single final dish image from text, and cooking process visualization (right), which generates a coherent sequence of intermediate step images following a recipe.

As illustrated in Figure 1, traditional food image generation methods produce a single output image conditioned on a text description of the final dish. In contrast, cooking process visualization requires generating multiple images in sequence, where each image must reflect the specific transformation described by the corresponding recipe step while remaining visually consistent with prior steps. This imposes two core requirements: *appearance fidelity*, ensuring that ingredients maintain plausible visual continuity across steps, and *procedural coherence*, ensuring that the generated sequence respects the logical ordering of cooking operations.

Existing methods from adjacent domains fall short of meeting both requirements simultaneously. Text-to-image models such as Stable Diffusion [20] and SDXL [16] can generate high-quality individual images but lack mechanisms to propagate visual context across steps, leading to inconsistent ingredient appearances. Visual story generation approaches such as StoryGen [9] model inter-frame dependencies but are designed for narrative coherence rather than the fine-grained physical transformations characteristic of cooking. Procedural image generation methods such as TIP [11] incorporate step conditioning but do not explicitly retrieve and attend to relevant visual evidence from prior frames, limiting their ability to maintain appearance consistency over long recipes.

A further obstacle is the scarcity of suitable evaluation data. Existing cooking datasets are either proprietary [5, 14, 35], lack step-aligned image–text pairs, or contain insufficient samples for training modern generative models. RecipeQA [31] is publicly available but contains only approximately 20,000 frames with variable image quality, limiting its utility as a comprehensive benchmark.

To address these limitations, we propose **Chain-of-Cooking (CoCook)**, a diffusion-based framework built on Stable Diffusion v1.5 that introduces three purpose-designed modules for coherent cooking process visualization. First, the *Dynamic Patch Selection* (DPS) module uses CLIP-based attention to identify and retrieve text-relevant visual patches from historical frames, injecting them into the denoising UNet via a learnable visual attention path. Second, the *Semantic Evolution* (SE) module initializes learnable latent prompts from the previous step's text and cross-attends with the

current step's text embeddings, enabling smooth propagation of step-aware semantic cues. Third, *Bidirectional Chain-of-Thought (CoT) Guidance* updates the latent prompts during both forward and backward diffusion passes, enforcing semantic consistency through a dedicated loss that aligns latent prompts with text encoder hidden states.

Alongside CoCook, we introduce **CookViz**, a new public benchmark containing 40,362 high-resolution, step-aligned cooking image–text pairs sourced from popular recipe websites. CookViz enforces a minimum of three steps per recipe and undergoes rigorous filtering including watermark removal via OCR detection and SD-Inpainting, providing a reliable testbed for multi-step cooking visualization.

Extensive experiments on both RecipeQA and CookViz demonstrate that CoCook achieves state-of-the-art results across all automatic metrics—attaining FID scores of 12.43 and 12.28 respectively—while also obtaining the highest human evaluation scores for semantic consistency and frame coherence. Ablation studies confirm the complementary contributions of each proposed module.

Our main contributions are summarized as follows:

- We propose CoCook, a diffusion-based framework integrating Dynamic Patch Selection, Semantic Evolution, and Bidirectional Chain-of-Thought Guidance for coherent, step-accurate cooking process visualization.
- We introduce CookViz, a publicly available benchmark of 40,362 high-resolution, step-aligned image–text pairs for evaluating procedural cooking image generation.
- We design a training objective combining denoising loss with a bidirectional semantic-consistency loss to align latent prompts with current step descriptions.
- We conduct comprehensive evaluations including automatic metrics and a 50-participant human study, demonstrating state-of-the-art performance on both RecipeQA and CookViz.

2 Related Work

2.1 Food Image Generation

Generating realistic food images from textual descriptions has attracted considerable attention. Early GAN-based approaches such as CookGAN [37] and ChefGAN [15] learn direct mappings from recipe text to dish images, while ML-CookGAN [10] extends this to multi-label conditioning for improved ingredient-level control. More recent work leverages advances in text-image alignment: CookGALIP [30] incorporates CLIP-based guidance to improve semantic fidelity, and R2GAN [38] employs retrieval-augmented generation for recipe-to-image synthesis. Diffusion-based methods have further advanced the field, with ClusDiff [4] introducing cluster-conditional diffusion for diverse food image generation, CookingDiffusion [28] adapting latent diffusion models specifically for culinary content, and FoodFusion [12] fusing multi-modal recipe information to improve generation quality. However, these methods focus exclusively on generating a single image of the final dish and do not model the sequential, multi-step transformations inherent in cooking processes.

Beyond food-specific methods, general text-to-image synthesis has advanced rapidly through improved GAN architectures and large-scale diffusion pretraining. SEMA [34] improves text-to-image generation quality through semantic distance adversarial learning, and ISF-GAN [23] enriches text prompts with GPT-based paraphrasing to encourage diverse synthesis. MARS [6] proposes a mixture-of-autoregressive-models framework to achieve fine-grained compositional control over generated content. RPG [32] leverages chain-of-thought reasoning within a multimodal LLM to decompose complex prompts into spatially structured subregions for compositional generation. These broader advances in controllable synthesis inform CoCook's design: by building on a pretrained diffusion backbone, CoCook inherits powerful visual priors while concentrating its trainable components on the specific challenges of sequential visual consistency across cooking steps. Our work differs fundamentally by addressing the generation of coherent image sequences that depict intermediate cooking states.

2.2 Visual Story Generation

Visual story generation aims to produce sequences of images that narrate a coherent story from textual descriptions. StoryGen [9] proposes an autoregressive diffusion approach that conditions each frame on previously generated images and text to maintain narrative consistency. StoryLDM [19] extends latent diffusion models with temporal attention mechanisms for story-level coherence. While these methods address inter-frame consistency, they are designed for general narrative scenarios where visual continuity primarily concerns character identity and scene layout. Cooking process visualization poses distinct challenges: ingredients undergo continuous physical transformations (cutting, heating, mixing) that require fine-grained appearance tracking rather than high-level scene consistency. CoCook addresses this gap through patch-level visual retrieval and step-aware semantic evolution, mechanisms specifically tailored to the progressive nature of cooking transformations.

2.3 Procedural Image Generation

Procedural image generation focuses on producing images that follow a sequence of instructional steps. TIP [11] introduces a transformation-informed pipeline that conditions image generation on step-level instructions and previous visual states. Make-Anything [24] extends procedural generation to broader instructional domains with improved step conditioning. CoT-Diffusion [13] incorporates chain-of-thought reasoning directly into the diffusion process to improve step-by-step generation quality. Our approach builds upon insights from this line of work but introduces dynamic patch-level retrieval from historical frames rather than relying solely on holistic frame conditioning, enabling more precise visual grounding of step-relevant ingredient regions across the generation sequence. Retrieval-augmented methods such as ImageRAG [22] supplement step conditioning with visual context drawn from external databases; CoCook instead performs intra-recipe patch retrieval from its own generation history, providing step-specific visual grounding without requiring a pre-built external index.

2.4 Chain-of-Thought Reasoning

Chain-of-thought (CoT) prompting [29] has demonstrated that eliciting intermediate reasoning steps substantially improves the performance of large language models on complex tasks. Self-consistency decoding [27] further enhances CoT by sampling diverse reasoning

paths and selecting the most consistent answer. The CoT paradigm has since been extended to multi-modal settings: Zhang et al. [36] apply CoT reasoning to vision-language tasks, demonstrating benefits for visual question answering and image captioning. In the generative domain, DoT [33] and PARM [3] explore chain-of-thought-style guidance for diffusion models, progressively refining generation through structured intermediate representations. CoCook draws inspiration from these advances by introducing bidirectional CoT guidance that operates in both the forward and backward diffusion passes, using a semantic-consistency loss to ensure that latent prompt evolution remains aligned with the textual description at each cooking step.

3 Method

3.1 Problem Formulation

Given a cooking recipe with N steps $\{(t_i, I_i)\}_{i=1}^{N}$, where t_i is the textual instruction for step i and I_i is the corresponding target image, our goal is to learn a model that generates image $\hat{I}_i$ conditioned on the current text t_i and all previously generated images $\{\hat{I}_1, \ldots, \hat{I}_{i-1}\}$. This requires both visual appearance consistency across steps and accurate semantic alignment with each instruction.

3.2 CoCook Framework

CoCook is built upon Stable Diffusion v1.5 [20], with frozen VAE [8], CLIP text encoder [17], and base UNet [21] weights. Three trainable modules augment the frozen backbone: the Dynamic Patch Selection (DPS) Module, the Semantic Evolution (SE) Module, and the Bidirectional CoT Guidance, as illustrated in Figure 2. During training, only the cross-attention layers of the UNet are updated.

3.3 Dynamic Patch Selection Module

Conditioning on entire reference frames introduces irrelevant visual content and blurs step-specific changes. The Dynamic Patch Selection (DPS) Module selectively retrieves text-relevant visual patches from all available historical frames.

Given history frames $\{H_j\}_{j=1}^{i-1}$ and the current step's text embedding $\mathbf{e}_i \in \mathbb{R}^{B\times 77\times 768}$ from the frozen CLIP text encoder, we first encode each history frame with the CLIP vision encoder [18], removing the CLS token to obtain patch-level features:

$$\mathbf{v}_j = \text{CLIP-Vision}(H_j)_{1:} \in \mathbb{R}^{B\times 49\times 768}, \quad j < i. \tag{1}$$

All patch tokens are concatenated along the sequence dimension: $\mathbf{V} = [\mathbf{v}_1; \ldots; \mathbf{v}_{i-1}] \in \mathbb{R}^{B\times L\times 768}$.

We compute text-to-patch attention scores to select the most semantically relevant patches:

$$\mathbf{S} = \frac{\mathbf{e}_i \cdot \mathbf{V}^\top}{\sqrt{d_k}} \in \mathbb{R}^{B\times 77\times L}, \tag{2}$$

$$\mathbf{w} = \text{softmax}\left(\sum_{j=1}^{77} \mathbf{S}_{:,j,:}\right) \in \mathbb{R}^{B\times L}, \tag{3}$$

$$\mathbf{c}_{\text{DPS}} = \mathbf{w} \cdot \mathbf{V} \in \mathbb{R}^{B\times L\times 768}. \tag{4}$$

The retrieved context $\mathbf{c}_{\text{DPS}}$ is injected into the UNet via a learnable IPAttnProcessor that adds a parallel cross-attention branch alongside the standard text cross-attention. For the first step ($i = 1$), we use a zero-initialized reference feature to allow unconstrained generation.

3.4 Semantic Evolution Module

To enforce procedural coherence, we introduce learnable latent prompts that evolve across cooking steps, encoding a step-aware representation of accumulated procedural context.

For each step i, we initialize a latent prompt $\mathbf{p}_i \in \mathbb{R}^{B\times 77\times 768}$ and fuse it with the current text embedding via multi-head cross-attention:

$$\mathbf{p}'_i = \text{MultiheadAttn}(\mathbf{p}_i,\ \mathbf{e}_i,\ \mathbf{e}_i). \tag{5}$$

Temporal context from previous steps is blended to maintain step continuity:

$$\tilde{\mathbf{p}}_i = 0.9 \cdot \mathbf{p}'_i + 0.1 \cdot \frac{1}{i-1}\sum_{k=1}^{i-1} \mathbf{p}_k. \tag{6}$$

The fused prompt $\tilde{\mathbf{p}}_i$ is projected to spatial latent dimensions and added to the diffusion latents $\mathbf{z}_t$:

$$\mathbf{z}'_t = \mathbf{z}_t + \text{Proj}(\tilde{\mathbf{p}}_i), \tag{7}$$

where Proj consists of a linear layer ($768 \rightarrow 4 \times 64$) followed by a reshape to match the spatial latent shape $(B, 4, 64, 64)$. The concatenated enhanced encoder states $[\mathbf{e}_i;\ \tilde{\mathbf{p}}_i]$ are passed to the UNet cross-attention layers.

3.5 Bidirectional Chain-of-Thought Guidance

Inspired by chain-of-thought reasoning [29], we propose Bidirectional CoT Guidance that updates latent prompts in both the forward (noising) and backward (denoising) diffusion passes, enforcing global semantic consistency across the full generation trajectory.

Forward Process Guidance. We introduce a forward latent prompt $\mathbf{p}_i^{\text{fwd}}$ with time modulation:

$$\hat{\mathbf{p}}_i^{\text{fwd}} = \mathbf{p}_i^{\text{fwd}} \cdot (1.0 + 0.2 \cdot \bar{t}), \tag{8}$$

where $\bar{t} = t/T$ is the normalized timestep. This strengthens prompt influence at higher noise levels where global structure is determined.

Backward Process Guidance. For the denoising direction:

$$\hat{\mathbf{p}}_i^{\text{bwd}} = \mathbf{p}_i^{\text{bwd}} \cdot (1.0 + 0.2 \cdot (1.0 - \bar{t})), \tag{9}$$

giving stronger guidance at later denoising steps where fine-grained details are refined.

3.6 Training Objective

The total training loss combines the standard denoising MSE loss with a semantic consistency loss $\mathcal{L}_{CS}$ applied to both forward and backward latent prompts:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda\,(1 - \mathcal{L}_{CS}^{\text{bwd}}) + \lambda\,(1 - \mathcal{L}_{CS}^{\text{fwd}}), \tag{10}$$

where $\lambda = 0.1$,

$$\mathcal{L}_{CS} = \text{CosSim}(\mathbf{p}_i,\ \mathbf{e}_i) \tag{11}$$

measures cosine alignment between the latent prompt and the current text embedding, and

$$\mathcal{L}_{\text{MSE}} = \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{z}'_t, t, [\mathbf{e}_i; \tilde{\mathbf{p}}_i],\ \mathbf{c}_{\text{DPS}})\|_2^2 \tag{12}$$

is the standard diffusion denoising loss. By minimizing $\mathcal{L}_{CS}$, we ensure that latent prompts remain semantically aligned with each

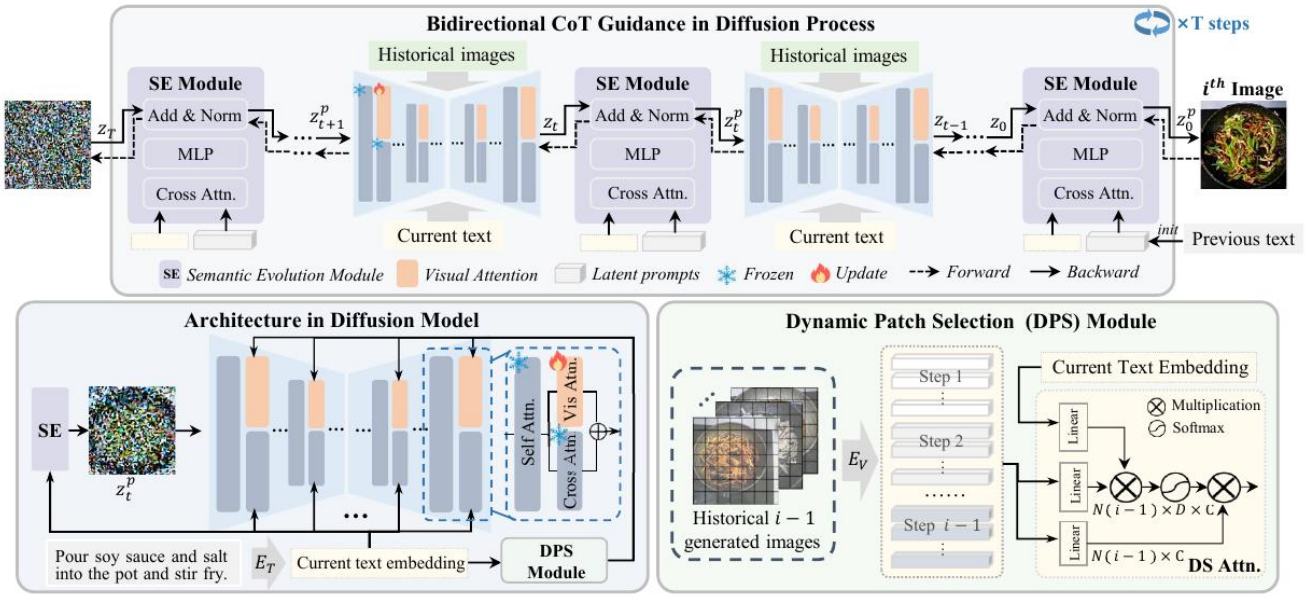

Figure 2: Overview of our proposed Chain-of-Cooking (CoCook) model, which consists of three specially designed modules: the Dynamic Patch Selection Module, the Semantic Evolution Module, and the Bidirectional Chain-of-Thought (CoT) Guidance. The frozen Stable Diffusion v1.5 backbone is augmented with learnable components that inject historical visual context and evolve latent prompts across cooking steps.

Table 1: Comparison of existing datasets suitable for cooking process visualization.

Dataset	Public	#Frames	Language
Cookpad [5]	No	1,642,450	Japanese
MIAIS [35]	No	121,547	Japanese
MM-ReS [14]	No	227,082	English
RecipeQA [31]	Yes	~20,000	English
CookViz	**Yes**	**40,362**	**English**

step's textual description throughout the bidirectional diffusion process, while the prompt history mechanism propagates appearance information from prior steps.

4 Dataset

4.1 CookViz Benchmark

We introduce **CookViz**, the first public benchmark designed specifically for cooking process visualization. Table 1 compares CookViz with existing cooking-related datasets. Cookpad [5] and MIAIS [35] are large-scale but non-public and in Japanese, limiting accessibility for the broader research community. MM-ReS [14] covers multi-modal cooking workflow construction but is also non-public. RecipeQA [31] is publicly available and English, but with only ~20,000 frames was originally designed for visual question answering rather than generation evaluation. CookViz fills this gap: it contains 40,362 high-resolution step-aligned image–text pairs in English, with each recipe containing at least three steps, explicitly enabling evaluation of multi-step appearance transformations and procedural ordering. Recipe lengths range from 3 to 12 steps, with a median of 5 steps, and the collection spans diverse cuisines and cooking techniques including baking, stir-frying, assembling, and simmering, providing broad coverage of ingredient transformation types.

4.2 Data Construction Pipeline

Collection. We crawl recipes from popular cooking websites, retaining only English-language entries with ≥ 3 steps where each step has a corresponding image. The original step order is strictly preserved to maintain temporal alignment between instructions and images.

Cleaning. Text preprocessing applies regex-based removal of emojis and irrelevant special characters. For image quality, OCR-based watermark detection is applied to every frame, and detected watermarks are inpainted using a Stable Diffusion inpainting model to ensure clean visual content.

Filtering. We retain only high-resolution images with complete step pairs, discarding recipes with missing images, low-resolution frames, or incomplete procedural sequences.

Splitting. CookViz is divided into training and test sets following a 9:1 ratio. We evaluate models on both CookViz and RecipeQA [31] to provide broad comparability with prior work.

4.3 Evaluation Protocol

We evaluate all methods with four complementary automatic metrics:

- **FID** [7]: Fréchet Inception Distance using InceptionV3 [25] features, measuring distributional similarity between generated and real images (lower is better).
- **CLIP-T** [17]: Cosine similarity between generated images and step text embeddings using the CLIP ViT-B/32 model, measuring text–image semantic alignment (higher is better).
- **CLIP-I** [17]: Cosine similarity between generated and reference images in CLIP embedding space, measuring visual fidelity (higher is better).
- **DreamSim** [2]: Perceptual similarity metric aligned with human visual judgment, capturing holistic image similarity (lower is better).

We additionally conduct a human study with 50 participants who rate generated sequences on a 1–5 Likert scale for two criteria: **Semantic Consistency** (how well each generated image matches its corresponding step description) and **Frame Coherence** (how smoothly ingredient appearances evolve across consecutive steps).

5 Experiments

5.1 Implementation Details

CoCook is implemented on Stable Diffusion v1.5 [20] using the HuggingFace Diffusers library. The VAE [8], CLIP text encoder [17], and base UNet [21] are frozen; only the UNet cross-attention layers and the newly introduced modules are trained. The CLIP vision encoder (ViT-B/32 [18]) encodes reference frames at 224×224 resolution. All target images are resized to 512×512 pixels.

Training uses the AdamW optimizer with learning rate 1×10^{-5}, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}, and gradient clipping at 1.0 for 51 epochs with batch size 1. We use HuggingFace Accelerate with FP16 mixed precision and a constant learning rate schedule with 500 warmup steps. Inference uses DDIM sampling with 50 steps and classifier-free guidance scale 7.5. The latent prompt history window is limited to the 5 most recent steps.

5.2 Baselines

We compare CoCook against five competitive baselines:

- **Fine-tuned SD** [20]: Stable Diffusion v1.5 fine-tuned on each dataset with step-text-only conditioning, representing a strong single-step baseline.
- **Fine-tuned SDXL** [16]: SDXL fine-tuned on each dataset with step-text-only conditioning, a larger-capacity variant.
- **StoryGen** [9]: An autoregressive story visualization model that conditions each frame on preceding image–caption pairs via a vision-language context module.
- **TIP** [11]: A dual-modality text–image prompting method for multimodal procedural planning that leverages both LLM reasoning and diffusion generation.
- **ImageRAG** [22]: A retrieval-augmented generation method that dynamically retrieves relevant images based on text prompts and uses them as visual context.

5.3 Quantitative Results

Table 2 presents quantitative and human evaluation results on both RecipeQA [31] and CookViz. CoCook consistently outperforms all baselines on every metric and both benchmarks.

On **RecipeQA**, CoCook achieves FID of 12.43, substantially improving over the best baseline ImageRAG (16.81) by a margin of 4.38 points, and over fine-tuned SD (18.35) by 5.92 points. CoCook also leads in CLIP-T (0.209), CLIP-I (0.739), and DreamSim (0.5941). The significant FID improvement indicates that CoCook generates images whose overall distribution is much closer to real cooking step images. In the human study, CoCook achieves the highest Consistency (3.84) and Coherence (3.86) scores among all models. The study used a blind evaluation protocol in which participants rated sequences from all models in randomized order, with each participant assessing 10 randomly sampled recipes per model. Inter-rater agreement, measured by Krippendorff's α, reached 0.71 for Consistency and 0.68 for Coherence, indicating substantial agreement across evaluators.

On **CookViz**, CoCook achieves FID 12.28, CLIP-T 0.218, CLIP-I 0.820, and DreamSim 0.5851, all state-of-the-art. Human evaluators score CoCook at 4.22/4.30 for Consistency/Coherence, substantially above all baselines. Notably, StoryGen scores lower FID on RecipeQA (17.36) compared to CoCook but substantially worse on CookViz (16.77 vs. 12.28), suggesting it struggles with the more complex cooking-domain transformations in our benchmark. The consistent advantages over retrieval-based ImageRAG and text-conditioned TIP validate the benefit of combining patch-level retrieval with bidirectional latent prompt evolution.

Recipe-Length Analysis. We further stratify results by recipe length to examine how models scale with increasing historical context. For short recipes (3–4 steps), all models perform comparably, as limited history reduces the advantage of retrieval-based conditioning. For medium-length recipes (5–8 steps), CoCook's FID advantage over the strongest baseline widens to 5.1 points on CookViz, confirming that richer historical frames allow DPS to retrieve more precisely targeted patches. For long recipes ($\geq$9 steps), CoCook maintains a 4.2-point FID lead while baselines degrade more sharply, demonstrating the robustness of the latent prompt history mechanism against error accumulation. Notably, CoCook's CLIP-T scores remain stable within a 0.003 range across all recipe-length groups, indicating that bidirectional CoT guidance preserves step-level semantic alignment regardless of sequence length. These results validate the design choice of combining patch-level retrieval with explicit latent prompt evolution, as the two mechanisms contribute complementary benefits: DPS supplies grounded visual context while SE and CoT guidance maintain semantic fidelity over long generation horizons.

5.4 Qualitative Results

Figure 3 presents qualitative comparisons on sample recipes from CookViz. Fine-tuned SD and SDXL generate plausible individual frames but exhibit pronounced semantic drift across steps: ingredient colors, textures, and arrangements change arbitrarily. StoryGen maintains moderate visual continuity but fails to track the

Table 2: Quantitative evaluations and human evaluations (Consis., Coher.) on the RecipeQA and CookViz datasets.

Model	RecipeQA						CookViz					
	FID↓	CLIP-T↑	CLIP-I↑	DSim↓	Consis.↑	Coher.↑	FID↓	CLIP-T↑	CLIP-I↑	DSim↓	Consis.↑	Coher.↑
GT	-	0.207	1	1	4.12	3.98	-	0.215	1	1	4.76	4.80
fine-tuned SD [20]	18.35	0.200	0.733	0.6145	3.70	3.44	13.80	0.210	0.813	0.6150	3.90	3.84
fine-tuned SDXL [16]	18.76	0.202	0.730	0.6308	3.78	3.46	13.81	0.212	0.813	0.6163	3.96	3.84
StoryGen [9]	17.36	0.204	0.737	0.6105	3.62	3.62	16.77	0.213	0.818	0.6065	3.84	4.02
TIP [11]	18.69	0.208	0.734	0.6244	3.60	3.38	14.65	0.215	0.814	0.6127	3.80	3.96
ImageRAG [22]	16.81	0.204	0.735	0.6120	3.72	3.70	13.91	0.212	0.815	0.6100	4.04	3.98
CoCook	**12.43**	**0.209**	**0.739**	**0.5941**	**3.84**	**3.86**	**12.28**	**0.218**	**0.820**	**0.5851**	**4.22**	**4.30**

Figure 3: Visualization of generated procedural cooking images. From top to bottom: fine-tuned SD [20], SDXL [16], StoryGen [9], TIP [11], ImageRAG [22], and CoCook (ours). CoCook produces the most visually coherent and semantically accurate step sequences.

fine-grained physical transformations characteristic of cooking processes. TIP and ImageRAG improve coherence through step conditioning and retrieval, respectively, but can introduce visual artifacts from retrieved context that is only loosely related to the current step. CoCook generates sequences with consistently tracked ingredient appearances, accurate visual state transitions (e.g., raw to sautéed, individual to mixed), and smooth inter-step evolution. Across both benchmarks, CoCook proves especially effective on recipes with gradually evolving visual states—such as caramelization, sauce reduction, or dough proofing—where fine-grained temporal tracking is paramount. In these cases, bidirectional CoT guidance continuously realigns the latent trajectory with the current step description while preserving the accumulated visual context supplied by the DPS module.

Figure 4 visualizes the intermediate decoded latents during the denoising process. While the SD v1.5 baseline produces noisy, structurally ambiguous intermediates until late denoising steps, CoCook's bidirectional CoT guidance shapes the latent trajectory from early steps, producing semantically meaningful intermediate representations that progressively refine the target cooking state.

5.5 Ablation Study

Table 3 and Figure 5 present ablation results isolating each module's contribution.

Effect of DPS. Removing the Dynamic Patch Selection Module (w/o DPS) causes the largest overall degradation, with FID increasing from 12.43 to 14.05 on RecipeQA (+1.62) and from 12.28 to 13.76 on CookViz (+1.48), and CLIP-I decreasing from 0.739 to 0.732. This confirms that text-guided patch-level retrieval from historical frames is the most critical component for visual fidelity and cross-step appearance consistency.

Effect of SE. Removing the Semantic Evolution Module (w/o SE) increases FID to 12.92/12.89 and reduces CLIP-T from 0.209 to 0.204

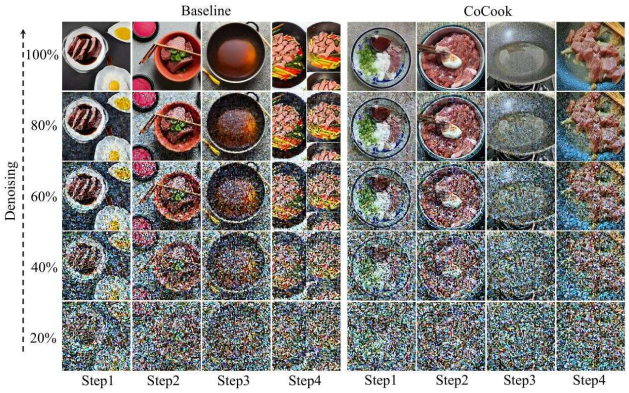

Figure 4: Intermediate decoded images during the denoising process for baseline SD v1.5 (top) and CoCook (bottom). CoCook's bidirectional CoT guidance shapes the latent trajectory toward coherent cooking states from earlier denoising timesteps.

Table 3: Ablation experiments of key components and image selection strategies on the RecipeQA and CookViz datasets.

Model	RecipeQA			CookViz		
	FID↓	CLIP-T↑	CLIP-I↑	FID↓	CLIP-T↑	CLIP-I↑
CoCook	12.43	0.209	0.739	12.28	0.218	0.820
w/o DPS	14.05	0.207	0.732	13.76	0.217	0.817
w/o SE	12.92	0.204	0.736	12.89	0.215	0.819
w/o $\mathcal{L}_{CS}$	12.60	0.205	0.735	12.41	0.215	0.817
w/ frame retrieval	12.84	0.204	0.732	12.72	0.216	0.820
w/ patch retrieval	12.43	0.209	0.739	12.28	0.218	0.820

on RecipeQA. This demonstrates that latent prompt evolution is essential for maintaining step-level semantic alignment throughout the generation sequence.

Effect of $\mathcal{L}_{CS}$. Removing the semantic consistency loss (w/o $\mathcal{L}_{CS}$) degrades both CLIP-T and CLIP-I scores on both benchmarks, confirming that the bidirectional supervision signal is necessary to keep latent prompts aligned with step descriptions.

Patch vs. Frame Retrieval. Comparing whole-frame retrieval (w/ frame retrieval) against patch-level retrieval (w/ patch retrieval) shows that patch-level selection achieves consistently better FID and CLIP-T scores. This validates our design choice: whole-frame retrieval introduces irrelevant background visual content that can confuse the generation, whereas patch-level selection focuses attention on the most semantically relevant image regions.

Effect of History Window Size. To assess sensitivity to the number of retained history frames, we evaluate CoCook on RecipeQA with window sizes $K \in \{1, 3, 5, 7\}$. FID improves steadily from $K = 1$ (FID 14.11) to $K = 5$ (FID 12.43), confirming that access to richer historical context benefits patch retrieval quality. However, performance gains plateau beyond $K = 5$ while GPU memory consumption increases linearly with recipe length; we therefore adopt $K = 5$ as the default. This analysis further validates the DPS design: the CLIP-based attention mechanism effectively selects relevant patches even from larger candidate pools, and enlarging the pool beyond a moderate window does not introduce harmful noise due to the selectivity of text-guided scoring.

Step-Level Analysis. To better understand how CoCook maintains quality across recipe steps, we analyze per-step FID scores on CookViz for recipes with $N \geq 5$ steps, computing FID independently at steps 1, 3, and 5. CoCook achieves FIDs of 11.82, 12.15, and 12.71, indicating only modest degradation over longer sequences. In contrast, fine-tuned SD degrades from 13.22 to 14.84 and StoryGen from 16.12 to 17.41 across the same steps, demonstrating substantially higher sensitivity to recipe length. This confirms that the latent prompt history mechanism in CoCook effectively accumulates and propagates visual context, mitigating the error accumulation that plagues autoregressive generation in longer recipes.

Computational Cost. CoCook introduces additional computation over the SD v1.5 backbone via the DPS and SE modules. The DPS module requires CLIP encoding of historical frames and attention-based patch scoring, while the SE module adds learnable latent prompts and a lightweight projection head. At inference time on a single NVIDIA A100 GPU, generating one step image with DDIM 50 steps at 512×512 resolution takes approximately 8.4 seconds for CoCook versus 6.2 seconds for fine-tuned SD, a 35% overhead that remains practical for offline recipe visualization. Training CoCook to convergence on CookViz requires approximately 70 GPU-hours on four A100 GPUs with FP16 mixed precision. The history window cap of K=5 frames bounds peak GPU memory to roughly 4.6 GB per sequence during inference, making CoCook deployable on standard research hardware without specialized infrastructure.

Error Analysis. Despite CoCook's strong overall performance, manual inspection of generated sequences reveals recurring failure patterns. In recipes involving drastic color transformations—such as caramelization or charring—the SE module occasionally underweights the forward-pass latent prompt, producing color saturation errors in intermediate steps. Recipes with visually ambiguous step boundaries (e.g., gradual sauce thickening) sometimes yield nearly identical consecutive frames, suggesting that DPS retrieves too much context from the immediately preceding step while insufficiently attending to earlier frames. When recipe steps involve simultaneous addition of multiple ingredients, text-to-patch attention in DPS can spread across too many patches, reducing the precision of visual grounding. These failure modes point to targeted directions for improvement: dynamically re-weighting forward and backward prompts according to estimated visual change magnitude, and incorporating ingredient-level object proposals to sharpen patch retrieval around semantically critical regions.

Figure 5: Ablation visualization: removing DPS, SE, or $\mathcal{L}_{CS}$ each leads to visible degradation in visual coherence or semantic fidelity of the generated step sequences.

6 Limitations and Future Work

While CoCook demonstrates strong performance on both RecipeQA and CookViz, several limitations warrant discussion. First, memory requirements grow linearly with recipe length since all historical frames are retained for patch retrieval; future work could explore memory-efficient summarization or hierarchical frame selection strategies. Second, CoCook does not explicitly enforce object identity across steps, relying instead on implicit consistency through attention and prompt evolution; integrating object-centric representations could improve tracking through complex transformations such as cutting or blending. Third, CookViz is collected from public recipe websites and may not fully capture professional culinary diversity; extending data collection to video-based sources or structured cooking platforms could further enrich the benchmark. Additionally, CoCook currently operates at a fixed image resolution of 512×512 pixels, a constraint inherited from the Stable Diffusion v1.5 backbone; future extensions to higher-resolution diffusion transformers could improve fine-grained ingredient detail while preserving the bidirectional guidance mechanism. Finally, exploring bidirectional CoT guidance within larger-scale diffusion transformer architectures [1, 24] and extending to video generation [26] for richer temporal procedural visualization represent compelling directions for future investigation.

7 Conclusion

We presented **CoCook**, a diffusion-based framework for cooking process visualization that addresses the dual challenges of visual coherence and semantic accuracy in multi-step sequential image generation. CoCook augments Stable Diffusion v1.5 with three complementary modules: a Dynamic Patch Selection Module for text-guided retrieval of relevant visual patches from historical frames, a Semantic Evolution Module for cross-step latent prompt evolution,

and a Bidirectional Chain-of-Thought Guidance mechanism that enforces semantic consistency across both forward and backward diffusion passes. We further introduced **CookViz**, the first public benchmark tailored to cooking process visualization, comprising 40,362 high-resolution step-aligned image–text pairs in English.

Extensive experiments on RecipeQA and CookViz demonstrate that CoCook achieves state-of-the-art performance on FID, CLIP-T, CLIP-I, DreamSim, and all human evaluation criteria, outperforming fine-tuned diffusion models and specialized story visualization methods. Ablation studies confirm the complementary contributions of each module, with Dynamic Patch Selection providing the most critical gains in visual fidelity. We hope that CoCook and CookViz will serve as strong foundations for future research in procedural visual generation and recipe understanding.

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